

# Bihar Engineering University, Patna

B.Tech 1<sup>st</sup> Semester Examination, 2024

Course: B.Tech

Code: 100114

Subject: Basic Electronics Engineering

Time: 03 Hours

Full Marks: 70

## Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

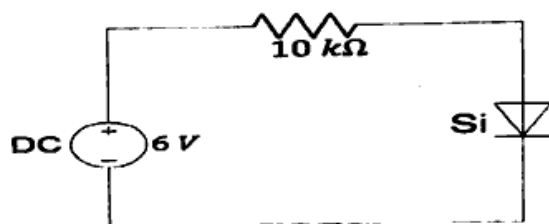
**Q.1** Choose the correct answer from the following (Any seven questions only):-

- (a) Which of these is used for amplification?  
(i) BJT in cutoff (ii) BJT in active mode  
(iii) BJT in saturation (iv) BJT in reverse active mode
- (b) Which terminal in JFET controls the current?  
(i) Gate (ii) Drain  
(iii) Source (iv) Substrate
- (c) BJT used as a switch operates in:  
(i) Active region (ii) Cutoff and Saturation  
(iii) Reverse active (iv) Linear region
- (d) What is the built-in potential typically across a silicon p-n junction?  
(i) 0.3 V (ii) 1.1 V  
(iii) 0.7 V (iv) 2.0 V
- (e) What is the peak inverse voltage for a center-tap full-wave rectifier?  
(i)  $V_m$  (ii)  $2V_m$   
(iii)  $V_m/2$  (iv)  $\sqrt{2} V_m$
- (f) Zener diode is mostly used in:  
(i) Amplification (ii) Voltage regulation  
(iii) Rectification (iv) Oscillation
- (g) In MOSFET, the gate terminal is:  
(i) Conductive (ii) Insulated from the channel  
(iii) Directly connected to the source (iv) Connected to the drain
- (h) The voltage gain of a voltage follower (buffer amplifier) is:  
(i) Zero (ii) Unity  
(iii) Infinity (iv) Negative
- (i) In 2's complement method, what is the 2's complement of binary number 0101?  
(i) 1011 (ii) 1101  
(iii) 1111 (iv) 1010
- (j) Which of the following is a universal gate?  
(i) AND (ii) NAND  
(iii) OR (iv) NOT

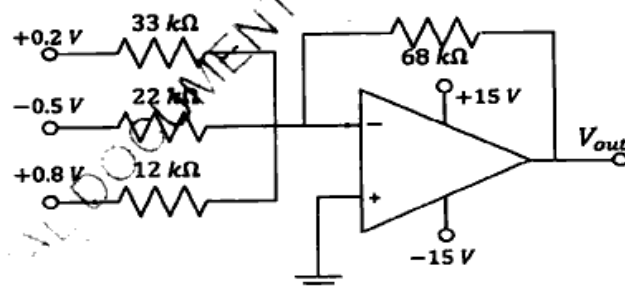
- Q.2** (a) Explain the construction and working of an Enhancement Type MOSFET and draw its characteristics. [7]
- (b) Define operating point. Explain the concept of load line analysis with suitable diagrams. [7]

- Q.3** (a) With a neat diagram, explain the working of a Zener diode as a voltage regulator and draw its V-I characteristics. [7]  
 (b) Describe the three configurations of BJT (CE, CB, CC) and their applications. [7]
- Q.4** (a) Derive the Shockley equation for JFET and explain the parameters involved. [7]  
 (b) Explain the difference between intrinsic and extrinsic semiconductors with suitable energy band diagrams. [7]

- Q.5** (a) Discuss the role of filter circuits in rectifier applications. Explain the capacitor filter with waveforms. [7]  
 (b) Determine the value of the current through the diode circuit given below. Assume  $V_f = 0.7\text{ V}$ . [7]



- Q.6** (a) Convert  $(0.513)_{10}$  to octal. [7]  
 (b) Explain how an Op-Amp integrator works and derive the output expression. [7]
- Q.7** (a) Discuss the mechanism of breakdown in a diode. Differentiate between Zener and Avalanche breakdown. [7]  
 (b) Find the output of the Op-amp. Assume Op-amp is ideal. [7]



- Q.8** (a) Explain the working of a Half Wave Rectifier and derive the expressions for average and RMS values of output voltage. [7]  
 (b) Implement an XOR gate using NAND gates. [7]
- Q.9** Write short notes on *any two* of the following: [7 x 2 = 14]  
 (a) Virtual ground in Op-Amp circuits.  
 (b) Describe the concept of thermal runaway in BJT.  
 (c) VI characteristics of LED  
 (d) Half- Adder